Quantum Random Access Coding and Its Applications

Harumichi Nishimura (Kyoto U.)

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Overview

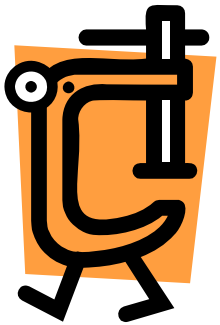
1. Quantum random access coding (QRAC)
   - QRAC for a few bits
   - Asymptotic bound
   - More bound of QRAC
2. Applications to advised computation
3. QRAC on the network

Quantum Random Access Coding

[Ambainis-Nayak-Ta-shma-Vazirani. 99]

(m,n,p)-QRAC

Sender



x

m bits

$|\Psi(x)\rangle$

n qubits



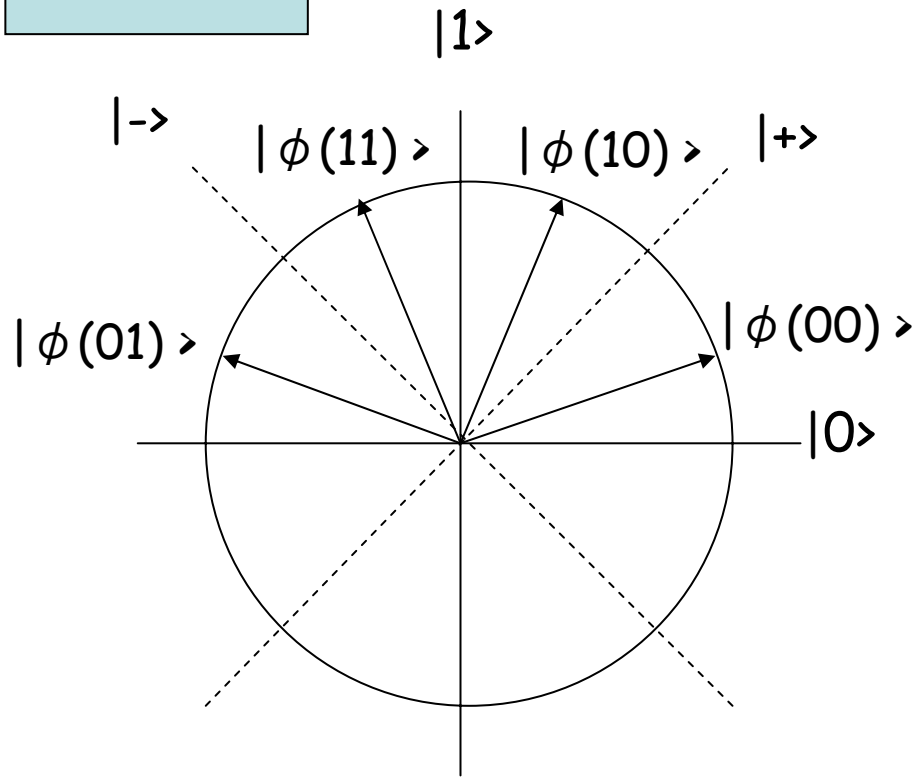
Receiver

$i \in \{1,2,..,m\}$

Goal: For any given i, output the i-th bit of x with prob. $p > 1/2$

(2,1,0.85)-QRAC

Encoding

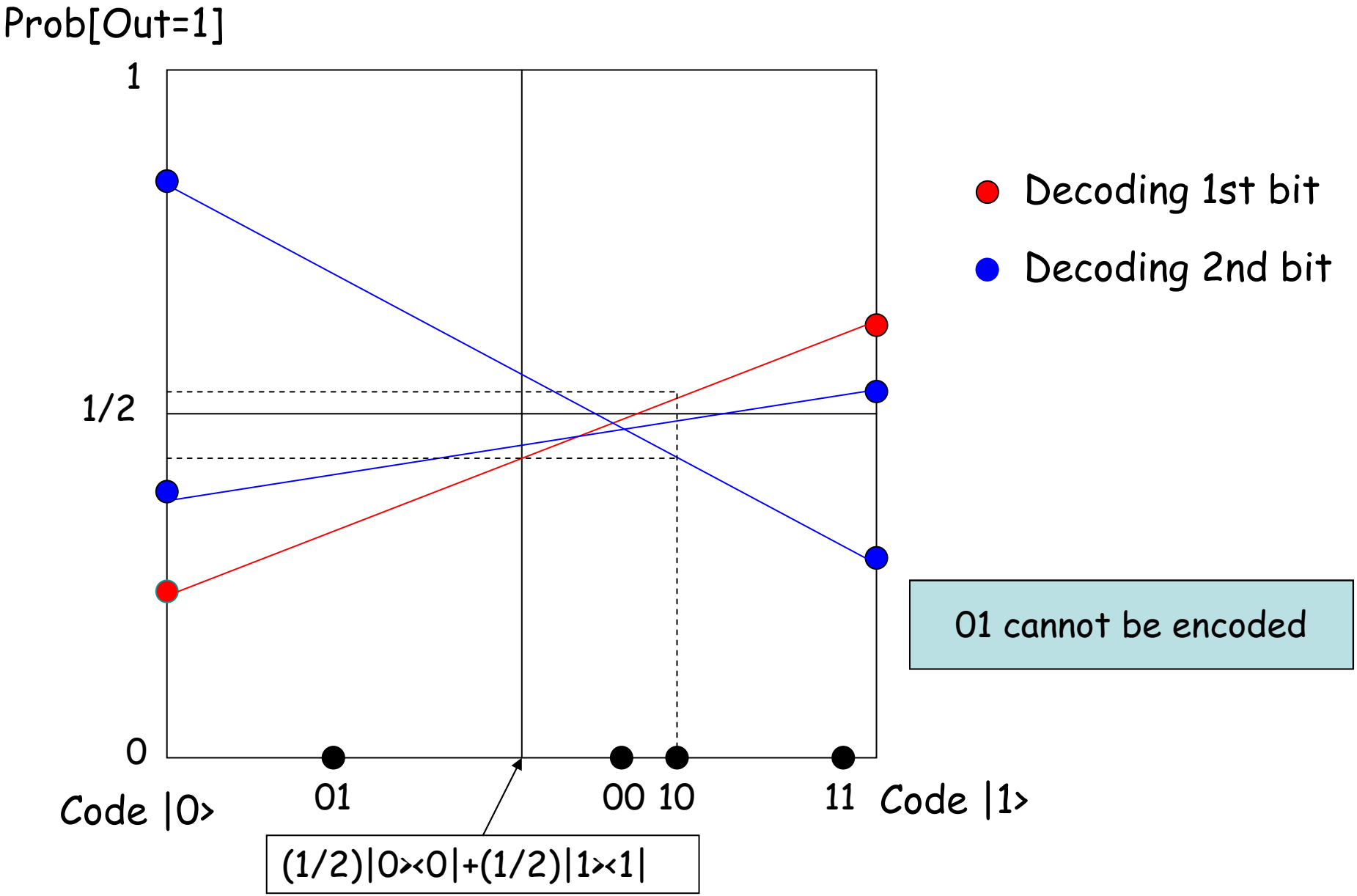


Decoding

To obtain 1st bit,
use the basis {|0>,|1>}.

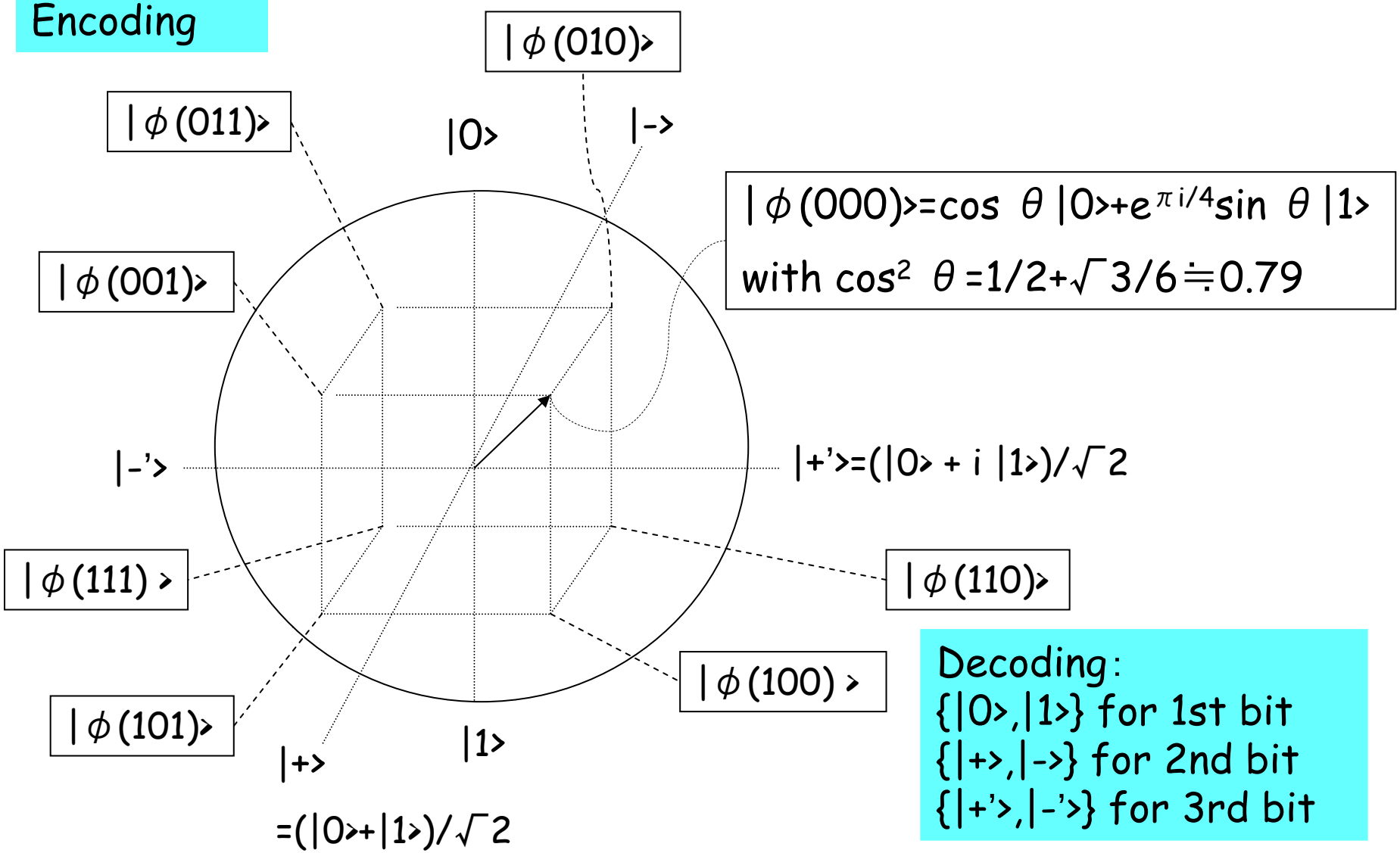
For 2nd bit,
use the basis {|+>,|->}

(2,1,>1/2)-RAC is Classically Impossible



(3,1,0.79)-QRAC

Encoding



Decoding:
{|0>,|1>} for 1st bit
{|+>,|->} for 2nd bit
{|+'>,|-'>} for 3rd bit

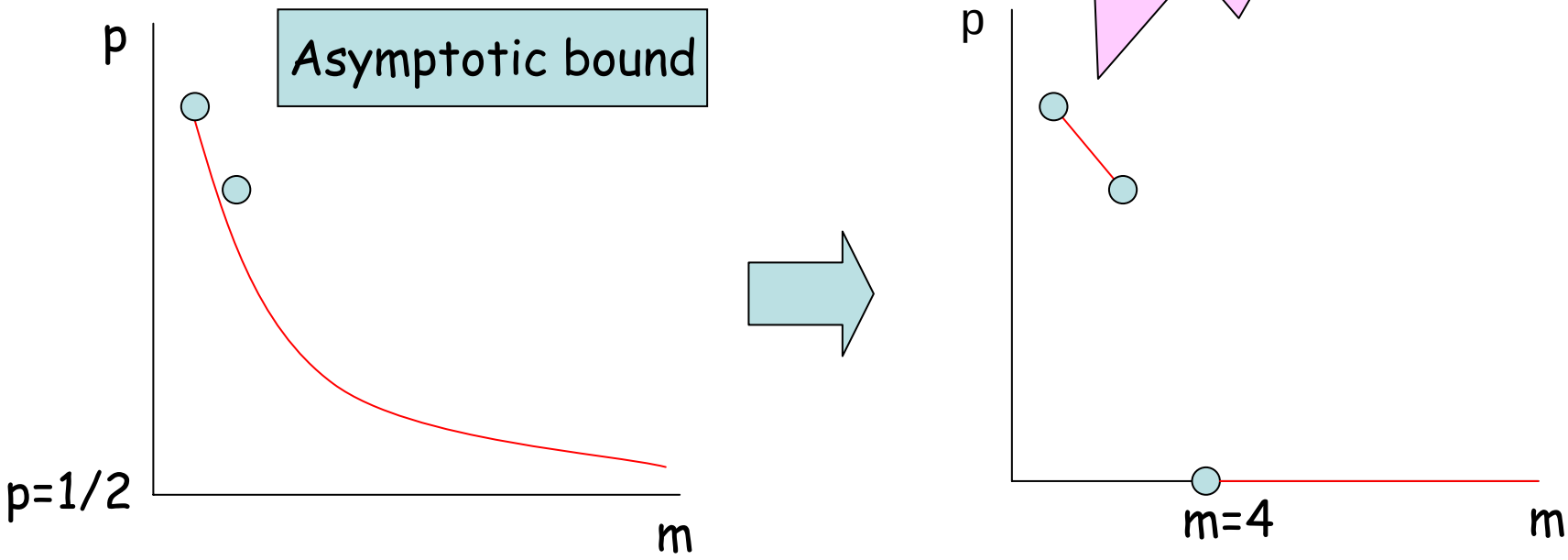
Asymptotic Bound

- Upper bound [Ambainis et al. 99]
  - (m,n,p)-QRAC exists such that
    $n \leq (1 - H(p))m + O(\log m)$
  - In fact, the coding is classical
  - Using "covering code"
- Lower bound [Nayak 99]
  - $n \geq (1 - H(p))m$
  - Proved by the use of quantum information theory
  - Many applications in computational complexity!

More Bound of QRAC

- Does an (m,1,p>1/2)-QRAC exis
  - (2,1,0.85)-QRAC
  - (3,1,0.79)-QRAC
  - (4,1,p)-QRAC, (5,1,p)-QRAC?

(4,1,p)-QRAC is impossible [Hayashi-Iwama -N-Raymond -Yamashita06]



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Applications of QRAC Lower Bound

- Limit of quantum finite automata
  [Ambainis et al. 99, Nayak 99]
- Communication complexity
  [Klauck 00, Buhrman-de Wolf 01, Gavinsky-Kempe-Regev-de Wolf. 06]
- Advised computation
  [N-Yamakami 04, Aaronson 05]
- Locally decodable code and private information retrieval
  [Kerenidis-de Wolf 03, Wehner-de Wolf 05]

Advised Computation

Input X (length n)



Advice Y_n



Adviser:
Unlimited power

Polynomial-time
Computer (P-machine)

He does not know her input x while knows its length n

P/poly := Class of sets recognized by P-machines with polynomial advice

Advised Quantum Computation

Input X (length n)



Advice Y
(length f(n))

Quantum
Advice $|\Phi\rangle$
(length f(n))



Adviser:
Unlimited power

Polynomial-time
Quantum Computer
(BQP-machine)

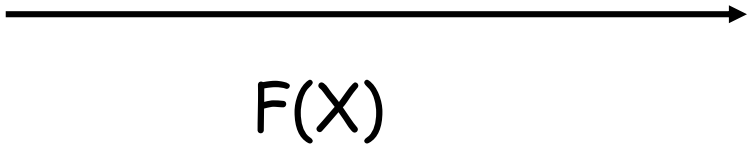
BQP/f(n) := Class of sets recognized by BQP-machines with advice of length f(n)

BQP/Qf(n) := Class of sets recognized by BQP-machines with quantum advice of length f(n)

Why Quantum Advice?

She sends messages securely using a quantum one-way function F





F(X)





Eve cannot find the message X

However, <u>if Eve has some quantum information</u> (say, by previous communication history), it can be quantum advice for Eve !

Limitation of Quantum Advice

- BQP/log$\subseteq$ BQP/Qlog$\subseteq$ BQP/poly$\subseteq$ BQP/Qpoly
- BQP/log $\neq$ BQP/Qlog
  - By using fingerprint
- For any positive function f, P/f(n) is not included in BQP/Q(0.08f(n)) [N-Yamakami 04]
  - By using the lower bound of QRAC
- Thus,
  - P/linear $\not\subseteq$ BQP/Qlog.
  - P/poly $\not\subseteq$ BQP/Qlinear, and hence
  - BQP/Qlog $\neq$ BQP/poly.

$P/f(n) \not\subseteq BQP/Q(0.08f(n))$

Recognized by P-machine + advice of length f(n)

Consider all the subsets L_n of $\{x \in \{0,1\}^n \mid x \leq s^{f(n)}\}$,

which have cardinality at most f(n).

BQP-machine M_n



Adviser

$|\Phi_n\rangle$: length m



$\text{Prob}[M_n(x, |\Phi_n\rangle) = L_n(x)] \geq 2/3$
for all $x \leq s^{f(n)}$ (= the f(n)-th string of $\{0,1\}^n$ in the lexicographic order).

The input x given to M_n is random for Adviser.

Thus, g(n) = $|\Phi_n\rangle$ is (f(n),m,2/3)-QRAC, and hence

some subset not recognized by M_n + quantum advice of length (1 – H(1/3))f(n) exists

QMA + Advice

- $L \in$ QMA/Qpoly $\Leftrightarrow$ $\exists$ ptime QC M, advice $\{|\Phi_n\rangle\}$

  If $x \in L$, then $\exists |\Psi\rangle$ Prob[$M(x,|\Psi\rangle,|\Phi_n\rangle)$="YES"] > 2/3

  If $x \notin L$, then $\forall |\Psi\rangle$ Prob[$M(x,|\Psi\rangle,|\Phi_n\rangle)$="YES"] < 1/3



Adviser

Advice $|\Phi_n\rangle$

Input X (length n)





Merlin (Could be malicious)

Witness $|\Psi\rangle$

Artur: BQP-machine

Limit of QMA + Advice

Input L (length $N=2^n$)



Adviser

Advice (length a)

Input X (length n)



Artur: BQP-machine



Merlin

Witness (length w)

This setting can be cosidered as (N, a, 1/3)-QRAC with witness of length w

- (De-Merlization) If ∃(N, a, 1/3)-QRAC with witness of length w, then ∃(N, $O(a \cdot w \log^2 w)$, 1/3)-QRAC. [Aaronson 05]
- QMA/Qpoly ⊆ PSPACE/poly

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Network Coding

- Network coding [Ahlswede-Cai-Li-Yeung. 2000]
  - Information theory + Transportation problem
  - Google "Network coding"

Information theory

- One-to-one
- Noisy channel





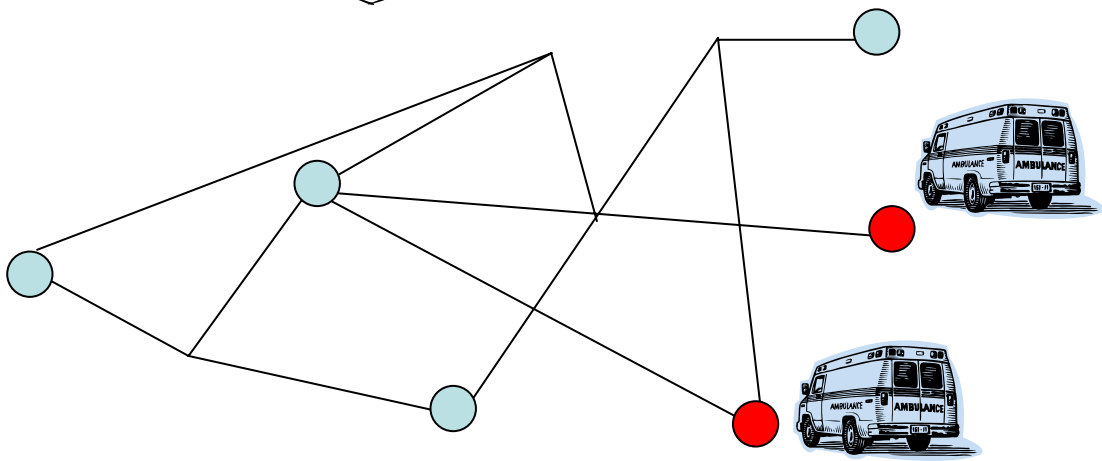
Data Compression

Error Correction

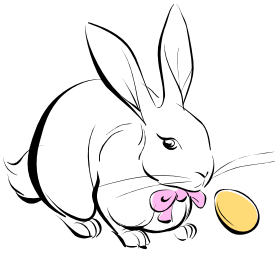
Security

Transportation problem

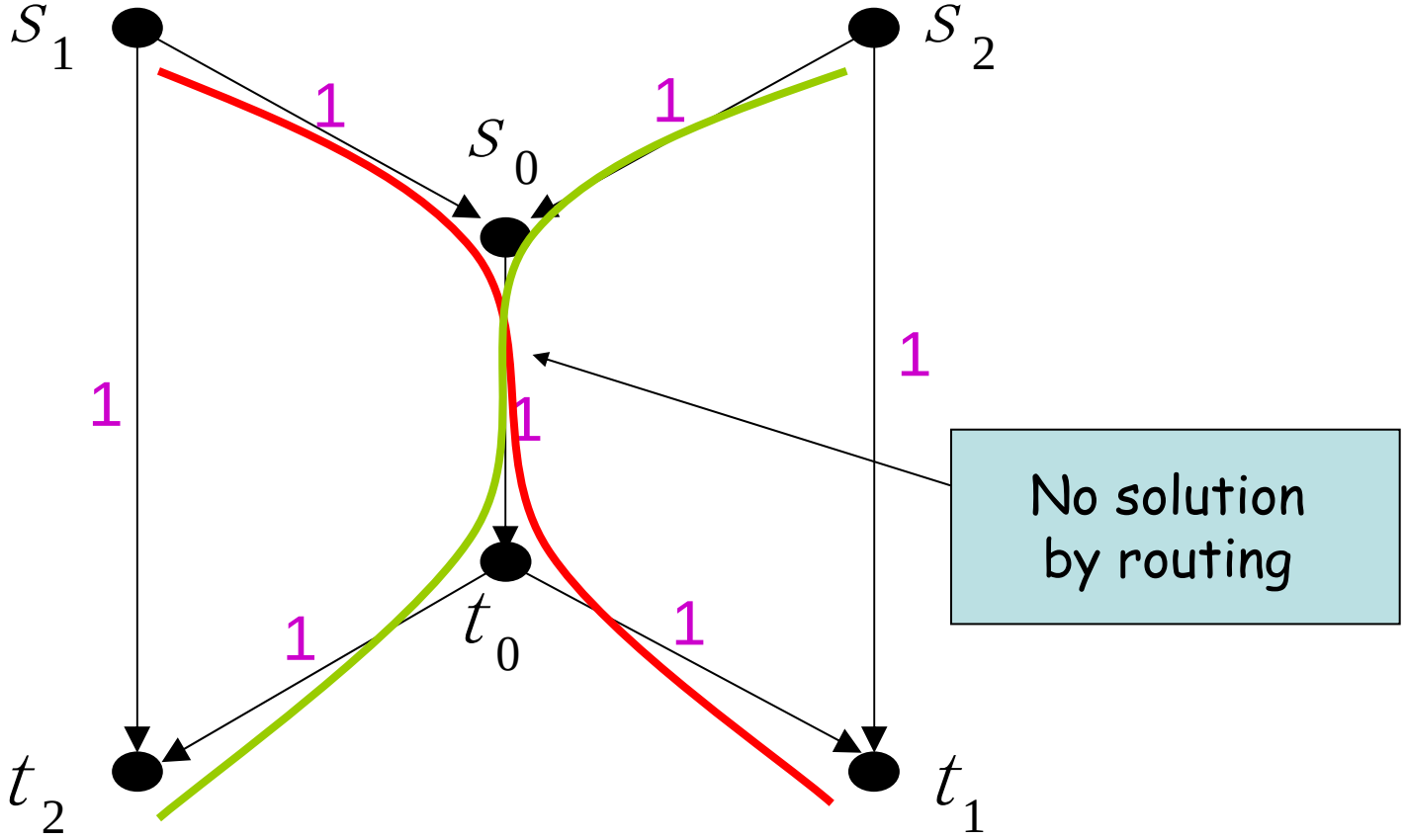
- Complicated network
- No coding



Liquid Flow



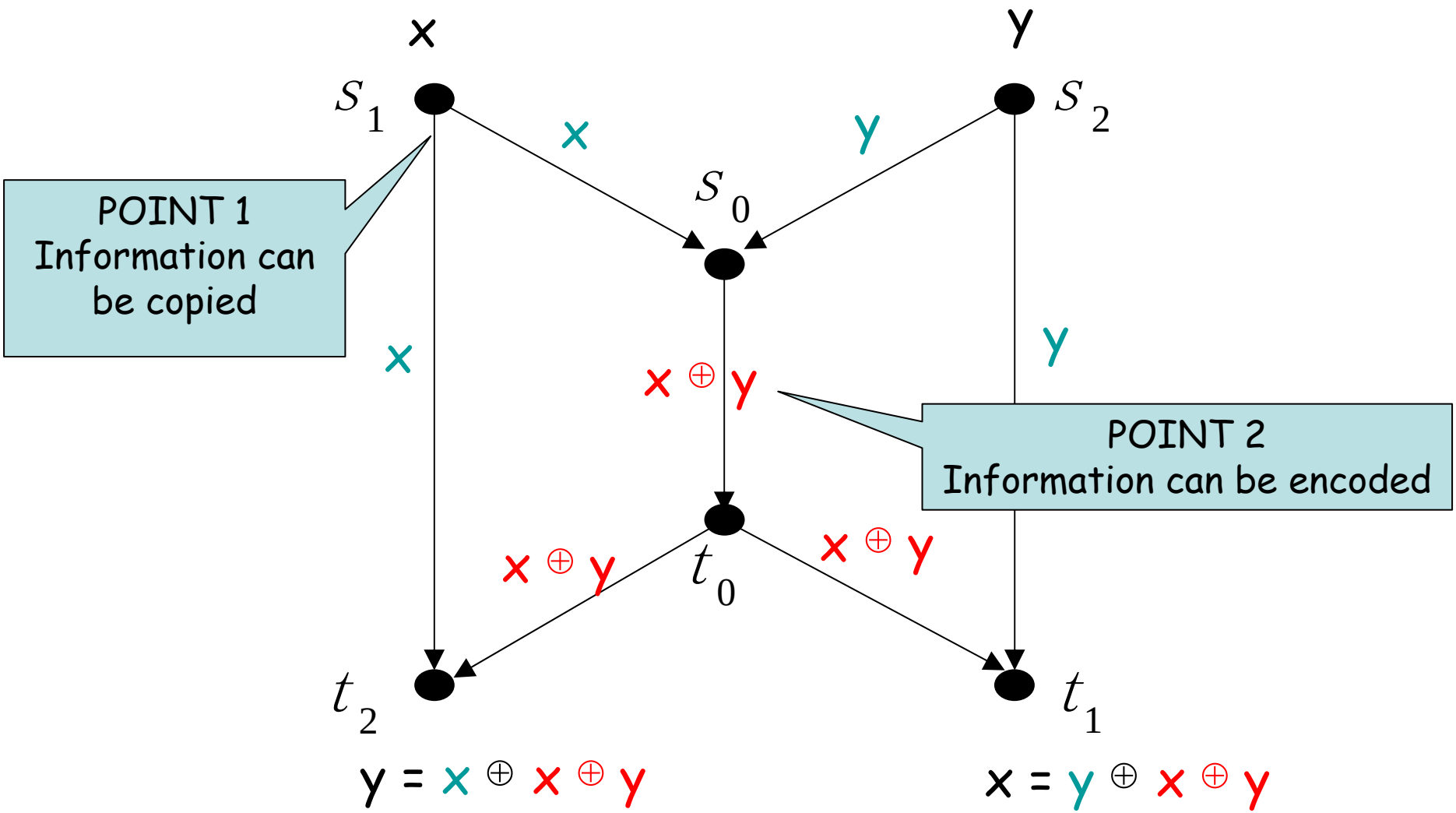




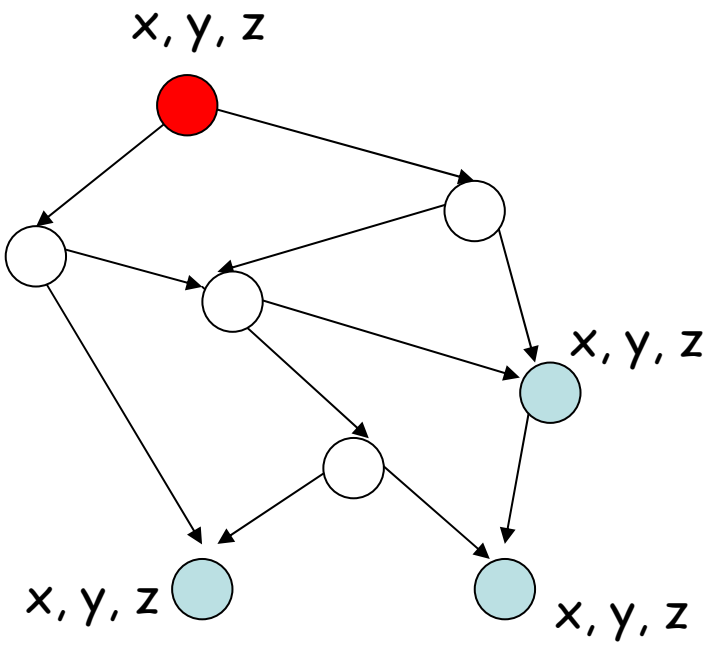
Butterfly Network (B-Net)

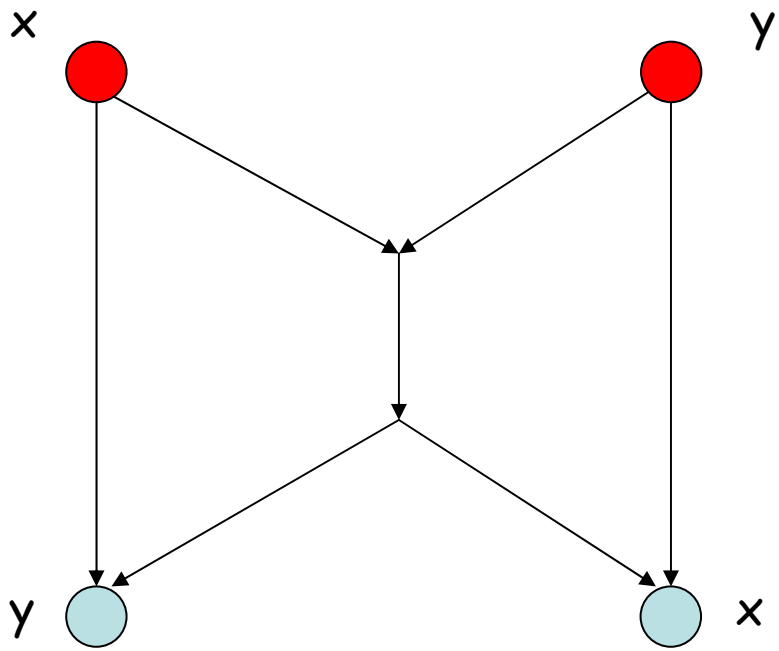
Information Flow

[Ahlswede et al. 2000]



Well-studied Network





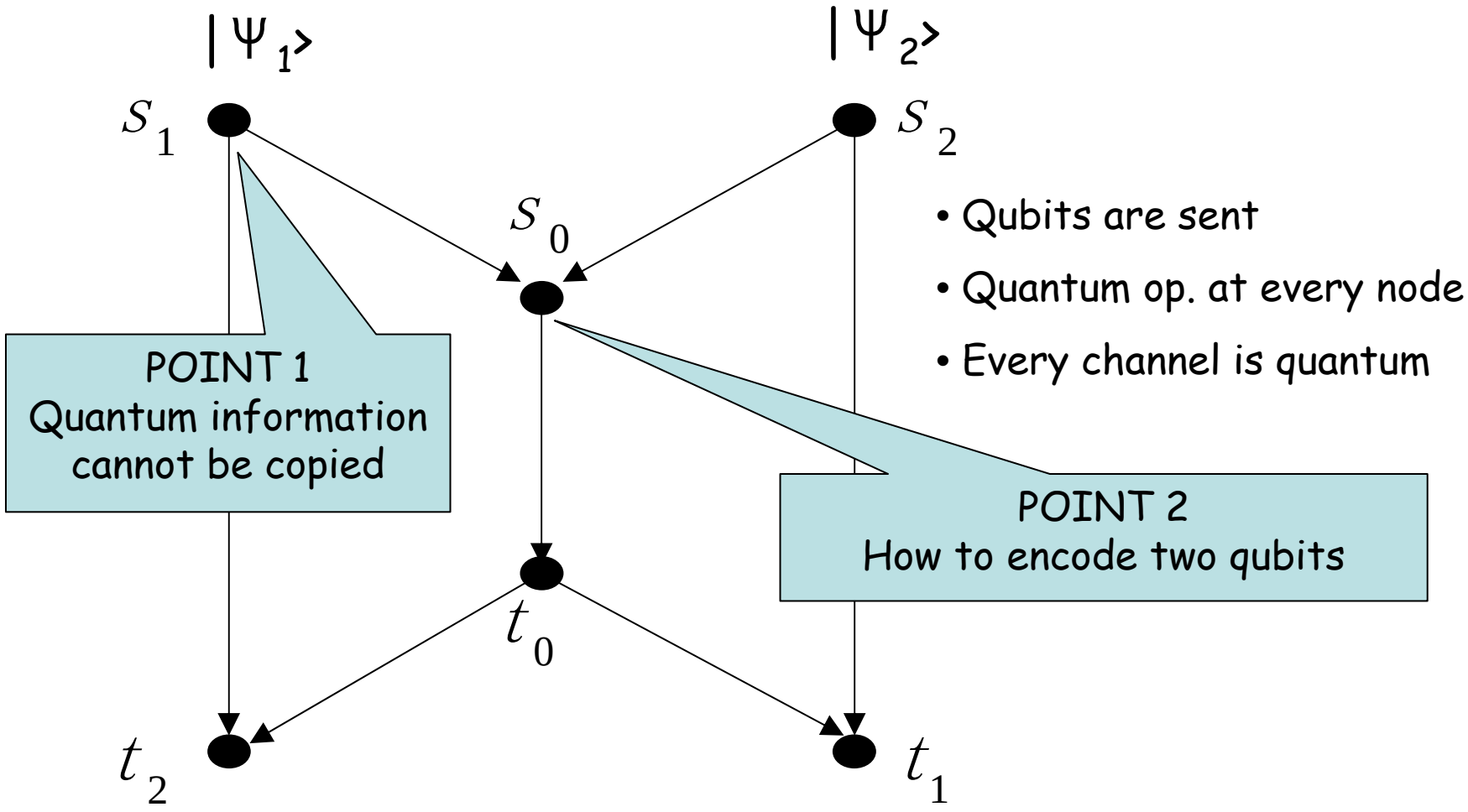
Multicast:

Every sink requires all messages which a source has.

k-Pairs Communication:

Each commodity has one source-sink pair. Butterfly network is 2-pairs communication problem.

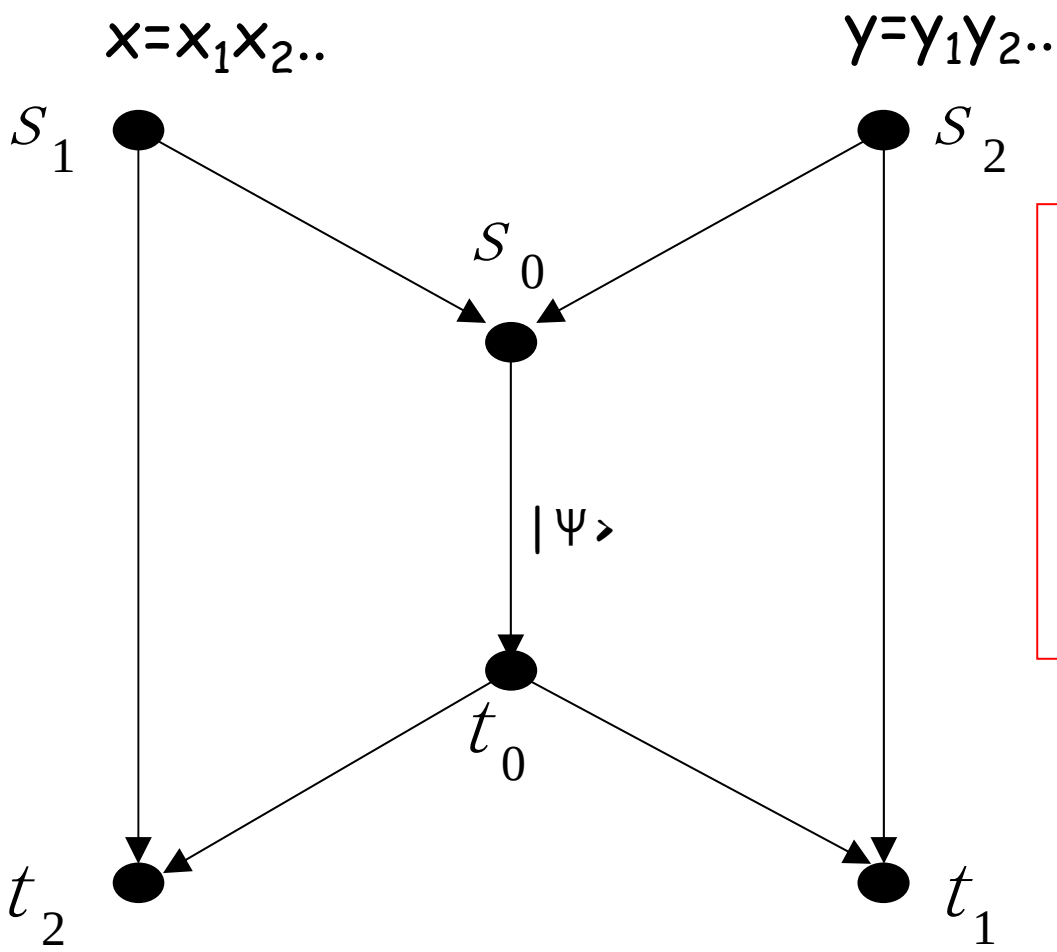
Quantum Information Flow



- Qubits are sent
- Quantum op. at every node
- Every channel is quantum

See quant-ph/0601088

Information Flow by Using Quantum

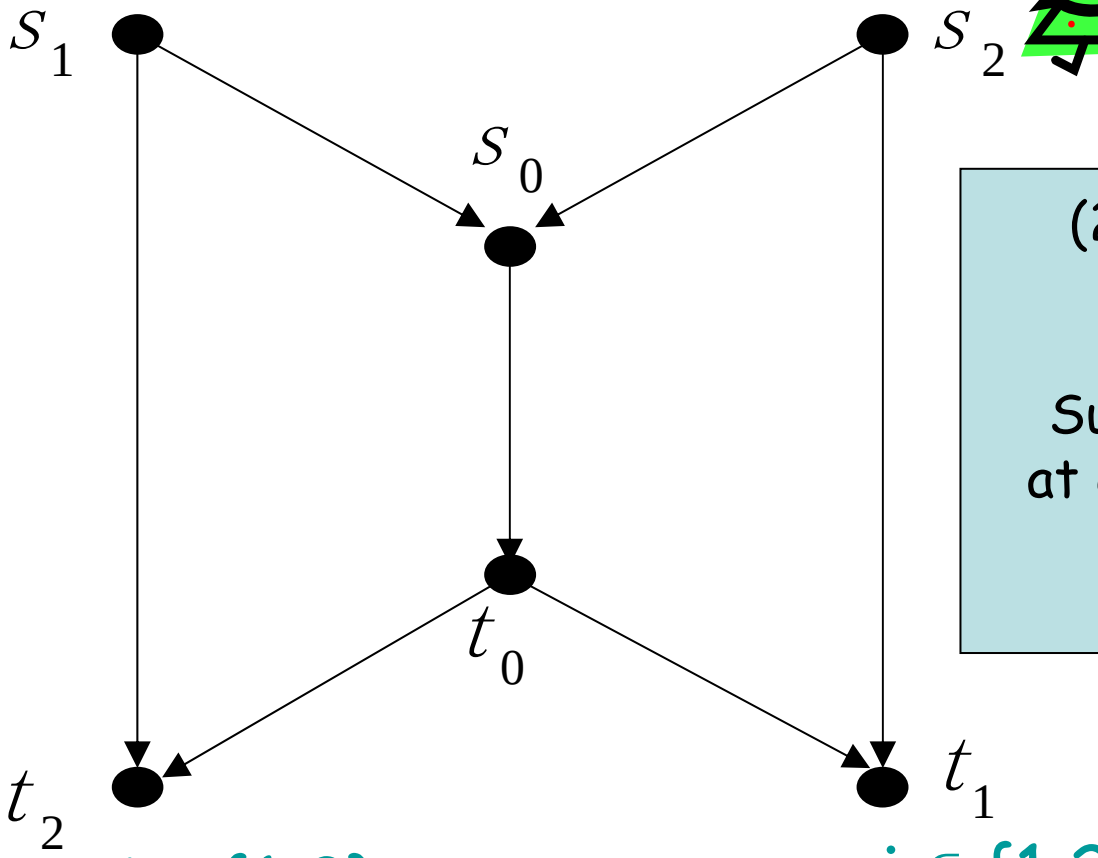


- **Many bits** are sent
- Problem: Success prob. of sending n bits by one-qubit is at most $1/2^n$
- Quantum op. at every node
- Quantum channel

QRAC on B-net











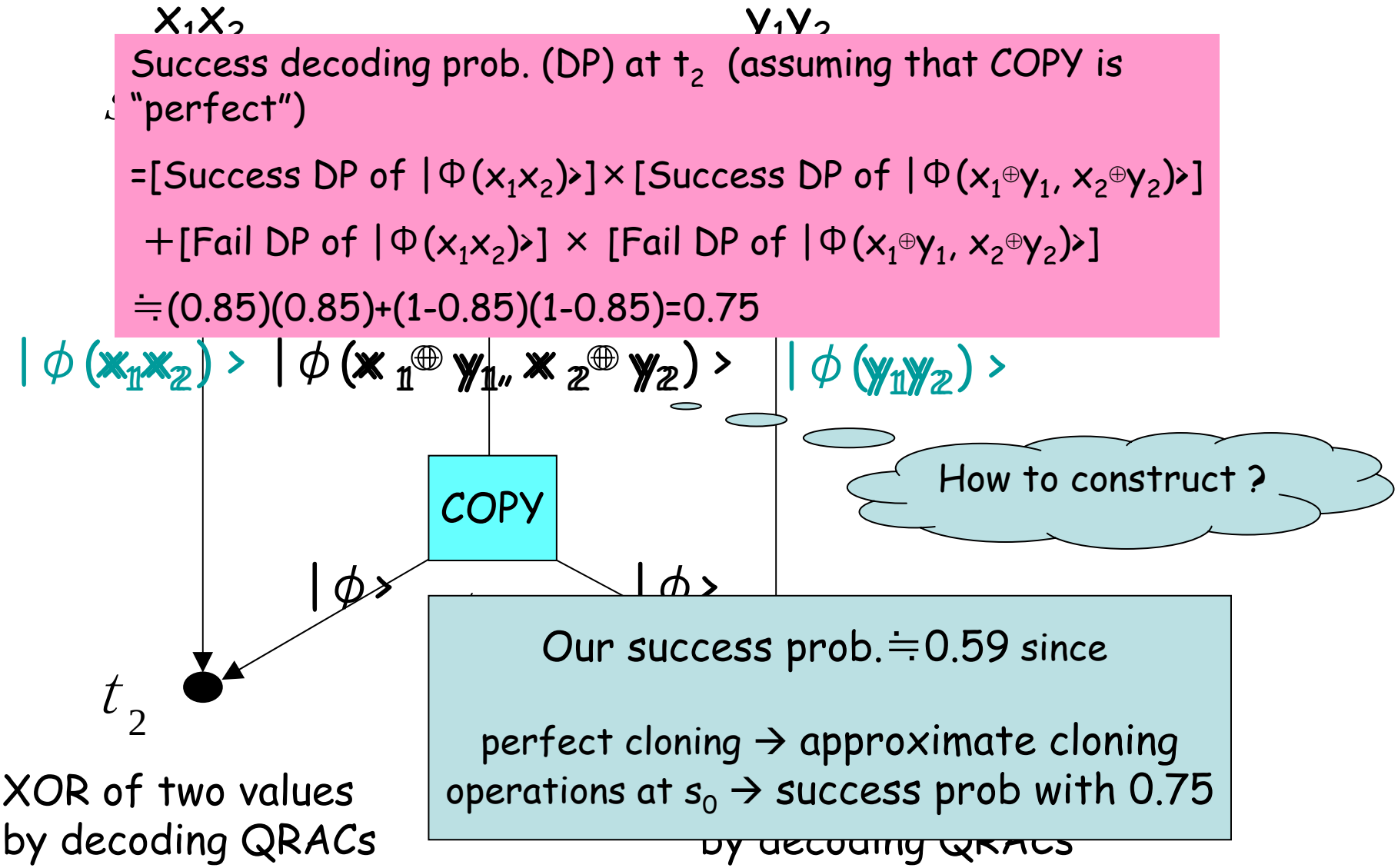
(2,1,p)-QRAC on B-net
(X2C2C)

Success decoding prob.
at each sink t is required
to be p > 1/2.

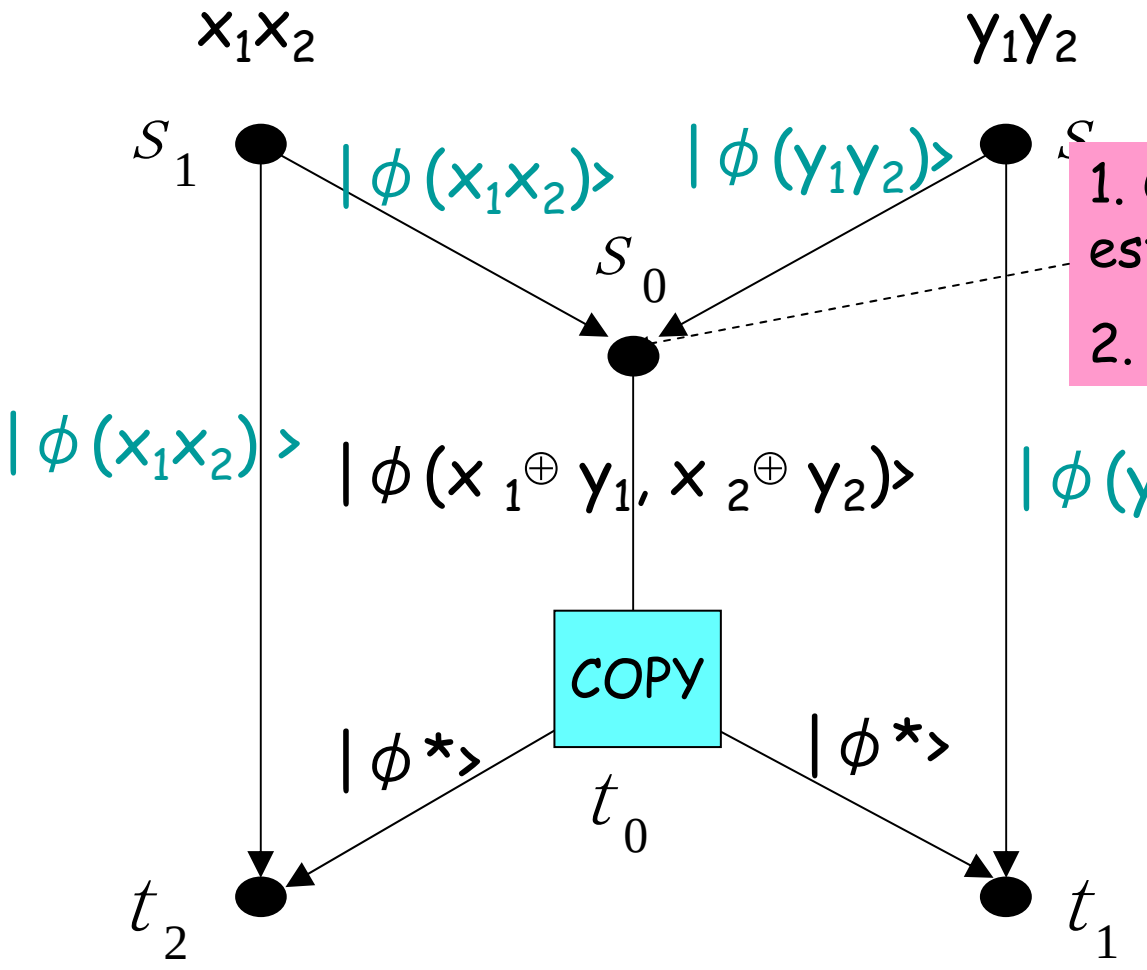
QRAC on B-net

- Success prob.≒0.59 for X2C2C on B-net [Hayashi-Iwama-N-Yamashita-Raymond 06]
  - Classically, at most 1/2
  - For the case of 3bits (X3C3C), success probability is still > 1/2 (≒0.525)
  - X4C4C is impossible

X2C2C


x_1x_2
y_1y_2
Success decoding prob. (DP) at t_2 (assuming that COPY is "perfect")
=[Success DP of $|\Phi(x_1x_2)\rangle$]×[Success DP of $|\Phi(x_1\oplus y_1, x_2\oplus y_2)\rangle$]
+[Fail DP of $|\Phi(x_1x_2)\rangle$] × [Fail DP of $|\Phi(x_1\oplus y_1, x_2\oplus y_2)\rangle$]
≒(0.85)(0.85)+(1-0.85)(1-0.85)=0.75
$|\phi(x_1x_2)\rangle$
$|\phi(x_1\oplus y_1, x_2\oplus y_2)\rangle$
$|\phi(y_1y_2)\rangle$
How to construct ?
COPY
$|\phi\rangle$
$|\phi\rangle$
t_2
Our success prob.≒0.59 since
perfect cloning → approximate cloning operations at s_0 → success prob with 0.75
XOR of two values by decoding QRACs
by decoding QRACs

X2C2C



Our method

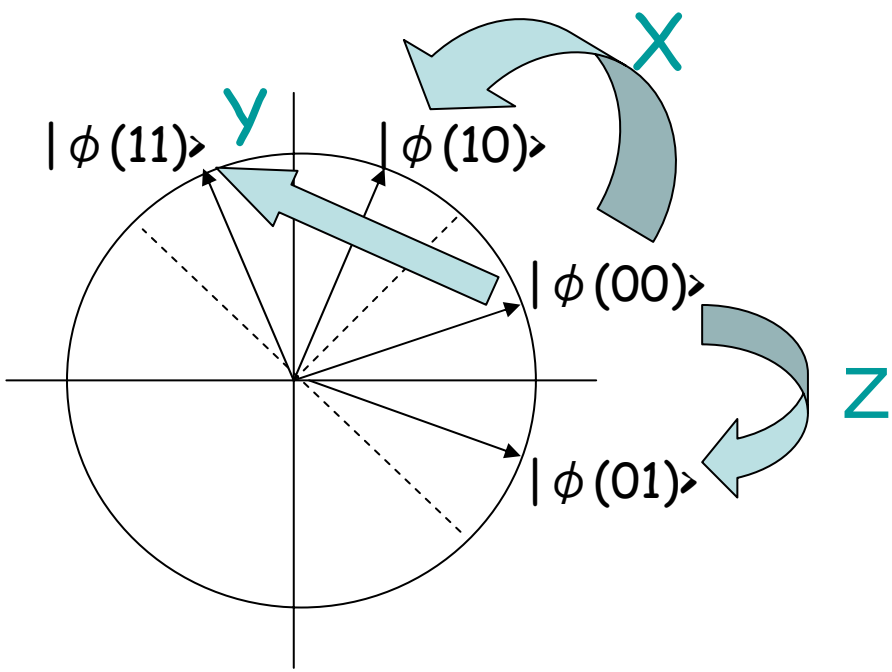
1. Obtain two bits z_1z_2 as an estimated value of y_1y_2
2. Construct $|\Phi(x_1 \oplus z_1, x_2 \oplus z_2)\rangle$

XOR of two values by decoding QRACs

XOR of two values by decoding QRACs

Operation at s_0 : If we know y_1y_2

$|\phi(x_1x_2)\rangle \rightarrow |\phi(x_1 \oplus y_1, x_2 \oplus y_2)\rangle$ is possible



Group operation

If y_1y_2=00,

Apply I= $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

If y_1y_2=01,

Apply Z= $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ phase flip

If y_1y_2=11,

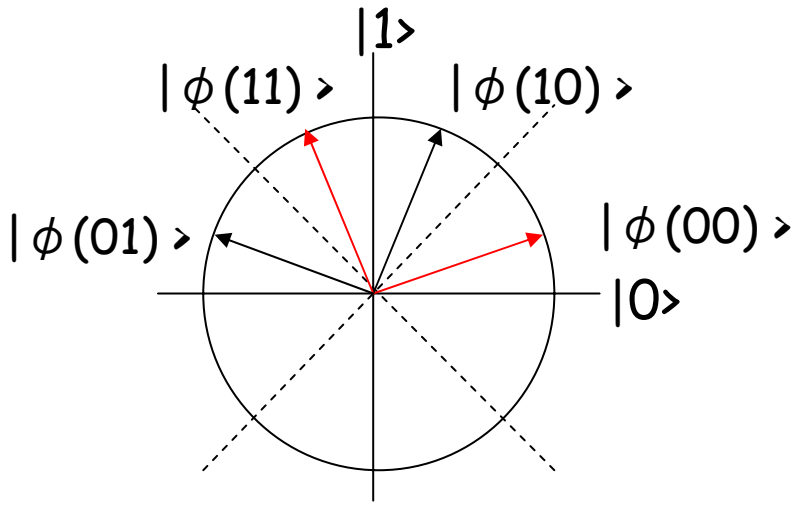
Apply Y= $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

If y_1y_2=10,

Apply X= $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ bit flip

Operation at s_0: Estimate y_1y_2

Random projective measurement MM_2



basis $\{|\phi(01)\rangle, |\phi(10)\rangle\}$

MM_2 = Choose 1 bit at random

1

If 0 is obtain, output 00

If 1 is obtain, output 11

$|\phi(10)\rangle \rightarrow MM_2 \rightarrow$

00 with prob. 1/4
01 with prob. 0
10 with prob. 1/2
11 with prob. 1/4

Conclusion

- Quantum random access coding has some advantage over classical one (even for on some network)
- Asymptotically, no advantage
  - (1 - H(p))m + O(log m) bits are enough for m bits
  - needs (1 - H(p))m qubits
- Including its variants, QRAC has many applications to computational complexity

